Questions

Q1.

Diagram 1 shows a glass prism which can be used to turn an image the right way up.

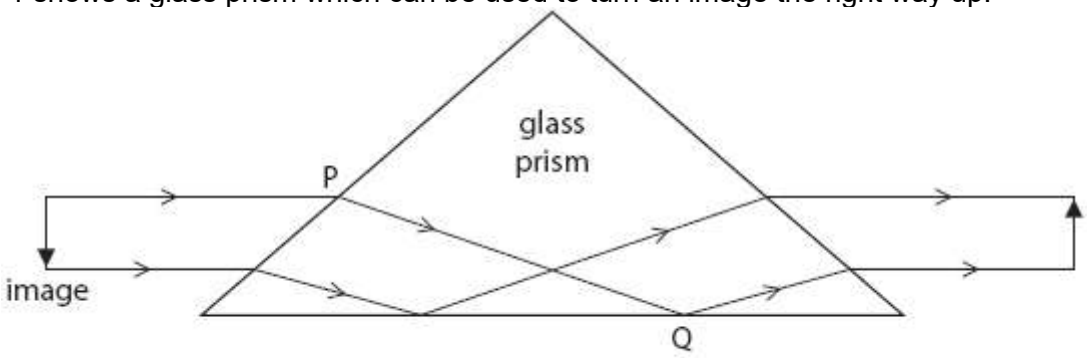


Diagram 1

(i) In diagram 1, total internal reflection occurs at Q.
Explain why total internal reflection occurs at Q.

(2)

.....

.....

.....

.....

(ii) The way in which the light changes direction at P is shown in diagram 2.
Mark on the diagram (*i*) for the angle of incidence and (*r*) for the angle of refraction for the ray of light shown.

(2)

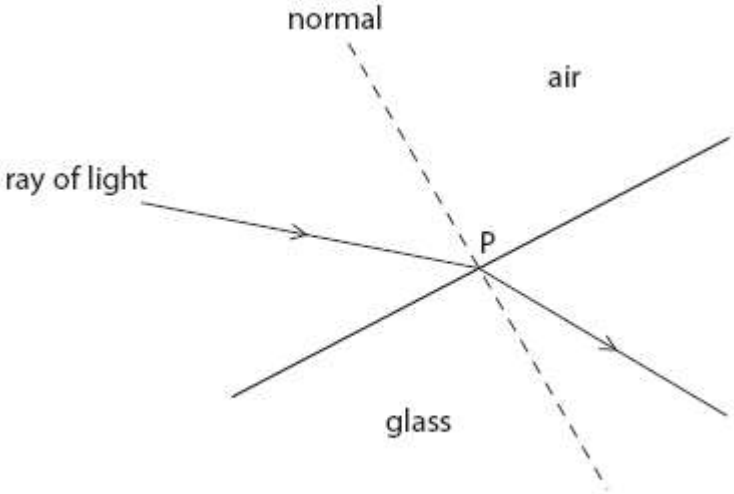


Diagram 2

(iii) Which of these is correct for the light as it enters the prism at P?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** frequency decreases

☐ **B** frequency increases

☐ **C** speed decreases

☐ **D** speed increases

Q2.

There is a piece of music called "The Flight of the Bumble Bee."
This takes 4 minutes to play.
During this time, a bee flies 1608 m.

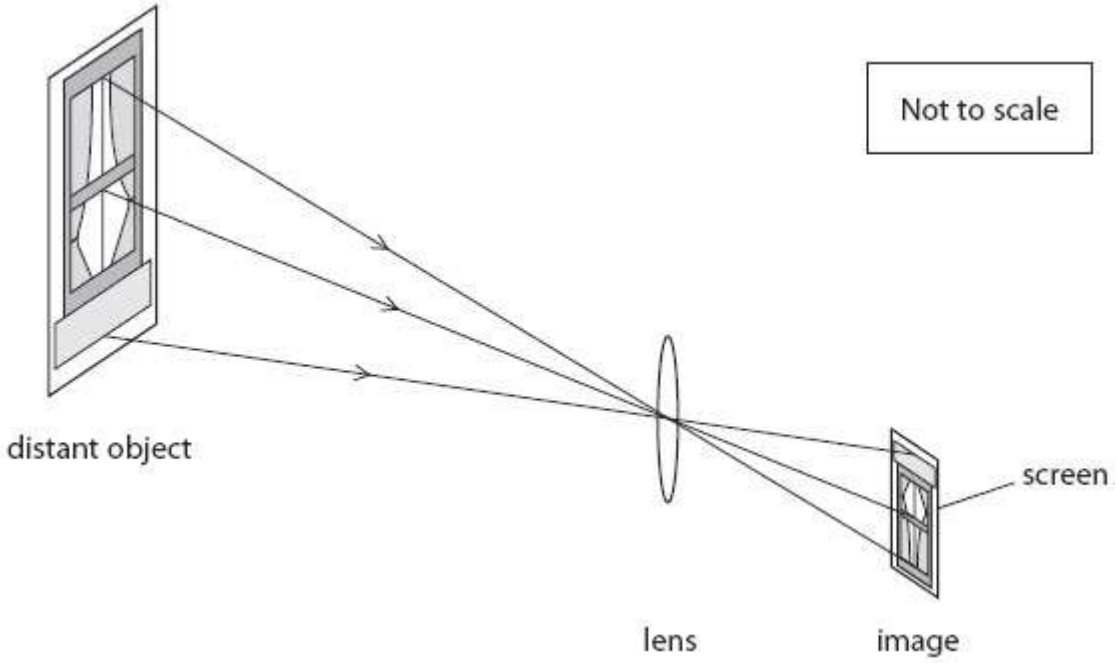
Calculate the average speed of the bee.

(3)

speed ..... m/s

Q3.

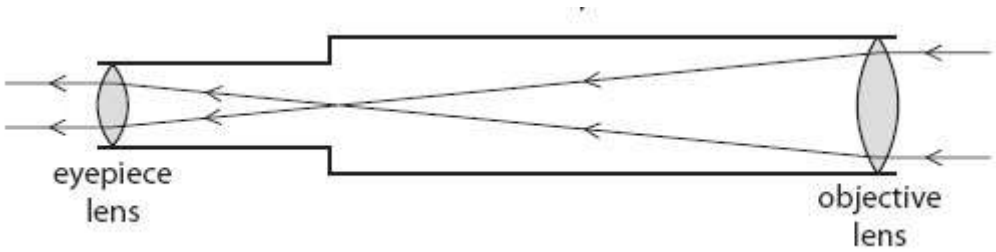
A lens can be used to produce a clear image of a distant object on a screen.



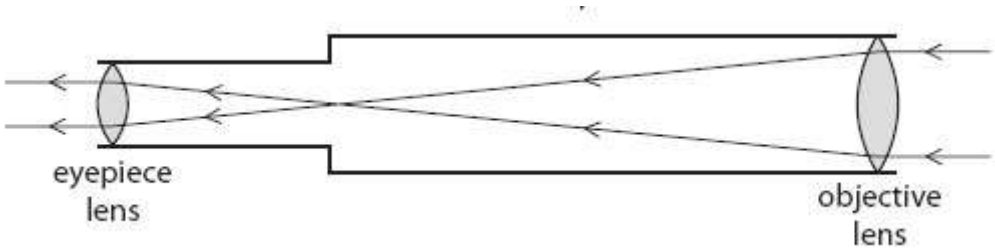
(a) (i) Complete the sentence by putting a cross (■) in the box next to your answer.

The image produced is real because it is

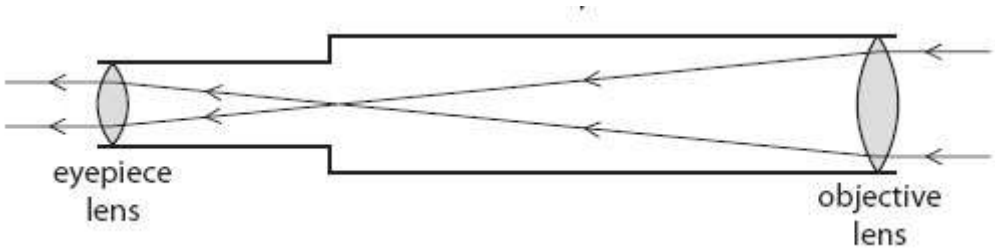
(1)



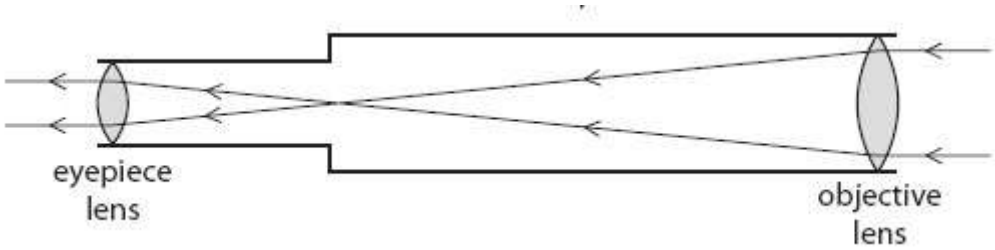
A in focus



B magnified



C on a screen



D smaller

(ii) Describe how to measure the focal length of the lens.

(2)

.....

.....

.....

.....

(b) The diagram shows a simple telescope which uses two lenses to look at stars.

(i) Explain what the eyepiece lens does.

(2)

.....

.....

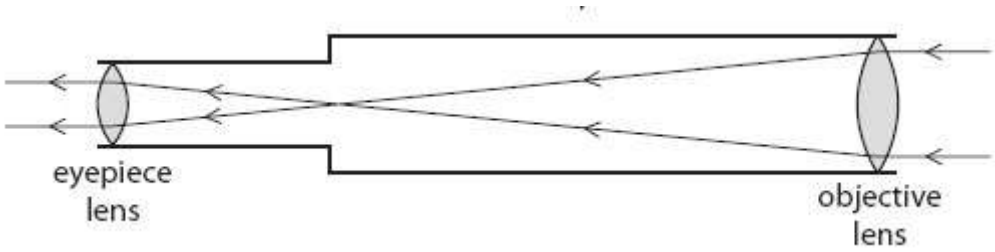
.....

.....

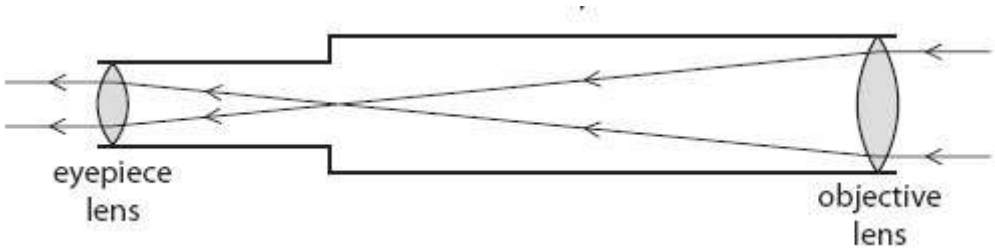
(ii) Complete the sentence by putting a cross (■) in the box next to your answer.

The light that travels from the stars transfers

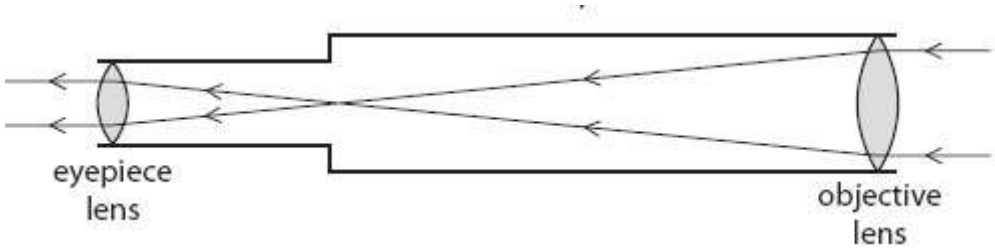
(1)



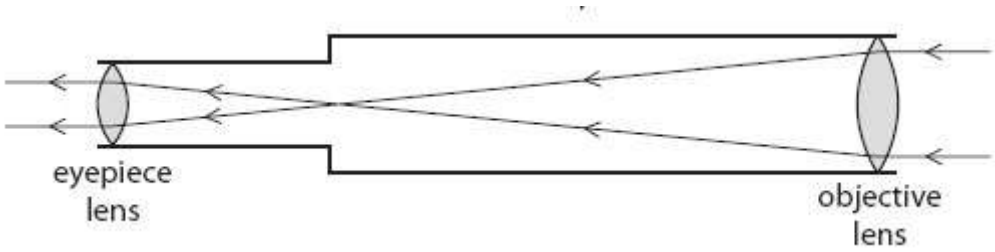
A charge



B energy



C mass



D matter

(c) Light and sound waves are produced at the same time by an explosion on Earth.

(i) The sound of the explosion is heard 1920 metres away 6.0 seconds after the explosion has happened.

Calculate the speed of sound in air.

(2)

.....

(ii) A scientist is standing a long way from the explosion.

Explain why he hears the explosion a few seconds after he sees it.

(2)

.....

.....

.....

.....

(Total for Question = 10 marks)

Q4.

Sound waves travel at 330 m/s in air.

A student sees a flash of lightning.

The student hears the sound of thunder 4.0 s later.

Calculate the distance from the student to the flash of lightning.

Use the equation

$$x = v \times t$$

(2)

distance = ... m

(Total for question = 2 marks)

Q5.

In a swimming pool, a wave is produced with a wavelength of 4.0 m and a velocity of 0.8 m / s.

Calculate the frequency of the wave.

State the unit of frequency.

(3)

Use the equation

$$v = f \times \lambda$$

frequency of wave ..... unit .....

(Total for question = 3 marks)

Q6.

The speed of a sound wave in air is 330 m/s.

The wavelength of this wave is 0.75 m.

Calculate the frequency of this wave.

Use the equation

$v = f \times \lambda$

(3)

frequency = ... Hz

(Total for question = 3 marks)

Q7.

The speed of sound in air is 300 m/s.

The speed of sound in water is 1500 m/s.

Calculate the ratio of the speed of sound in air to the speed of sound in water.

(2)

ratio of speed of sound in air to the speed of sound in water = ...

(Total for question = 2 marks)

Q8.

Blue light has a wavelength of 470 nm and a frequency of 6.30×10^{14} Hz

Calculate the velocity of blue light.

(2)

velocity = ... m/s

(Total for question = 2 marks)

Q9.

A water wave has a wavelength of 0.25 m and a frequency of 1.5 Hz.

Calculate the wave speed.

(2)

wave speed = ... m/s

(Total for question = 2 marks)

Q10.

Figure 12 shows part of a wave.

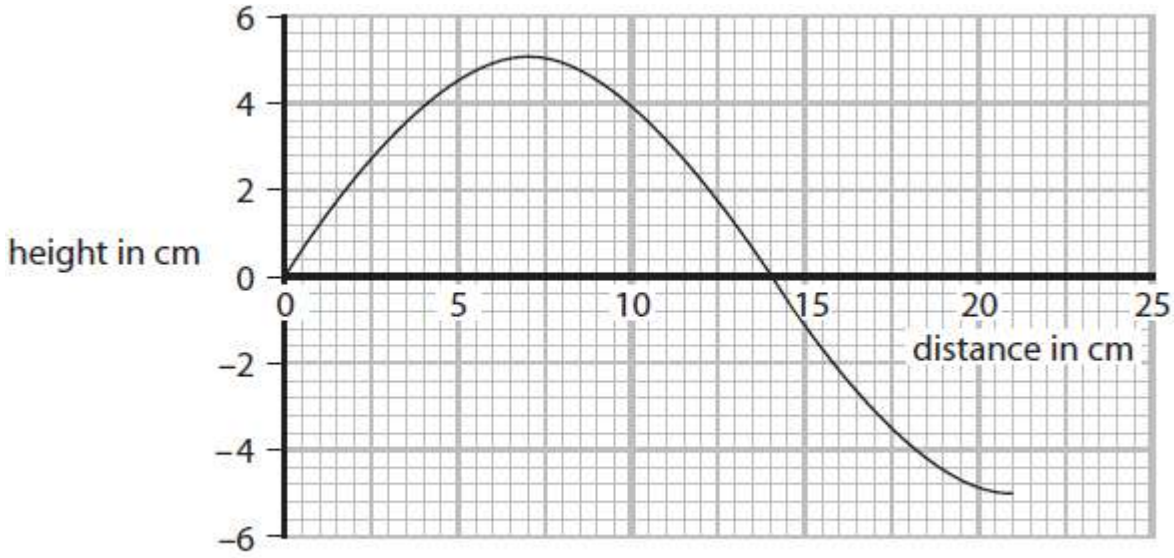


Figure 12

Use data from Figure 12 to calculate the wavelength of the wave.

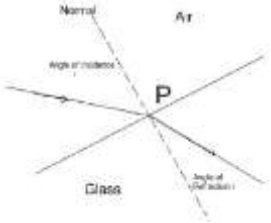
(2)

wavelength = ... cm

(Total for question = 2 marks)

Mark Scheme

Q1.

	Answer	Acceptable answers	Mark
(i)	An explanation linking: Angle (of incidence) in glass (1) greater than critical angle / 42° (1)	Angle in air cannot be greater than 90° for 1 mark Glass has a higher refractive index than air for 1 mark	**(2)**
(ii)	 angle i from normal in air (1) angle r from normal in glass (1)	accept for 1 mark angle i in air <u>and</u> angle r in glass/ <u>both</u> angles measured from normal	**(2)**
(iii)	☒ **C** speed decreases		**(1)**

Q2.

Question Number	Answer	Acceptable answers	Mark
	conversion of time 4x60 (1) substitution (1) 1608 / (4x60) **ecf if conversion shown** evaluation (1) 6.7 (m/s)	award full marks for correct answer with no working [1608 / 4 for 1 mark for these two] **allow** 402 for 2 marks **accept** for 2 marks: 5.36 (t=300 s 60→120→180→240→300, i.e. 4 steps of 60) 4.02 (t=400 s based on the misconception of 100 s to 1 minute) **allow** maximum of 1 mark for any other power of 10 error if no working	**(3)**

Q3.

	Answer	**Acceptable answers**	**Mark**
(a)(i)	☒ C on a screen		**(1)**
(a)(ii)	Explanation linking the following:- (measure) distance / length / from / line (1) lens to image / screen / focal point (1)	how far away use ruler / tape measure measure u(object distance) and v(image distance) calculate focal length using 1/u + 1/v = 1/f (measure) from lens to image (2)	**(2)**
(b)(i)	A description including the following: • magnifies • the image • refracts the light	brings nearer / zooms in / looks closer / makes bigger / enlarges intermediate / real image	**(2)**
(b)(ii)	☒ B energy		**(1)**

(c)(i)	Substitution v = 1920/6.0 (1) Evaluation (1) 320 (m/s)	300 (m/s) give full marks for correct answer, no working (2) Power of 10 error max. 1 mark.	**(2)**
(c)(ii)	Suggestions including the following: • recognition of any difference in speed / velocity (1) • correct difference in speed (1)	e.g. sound travels faster / quicker than light (1) c>v / v<c / c>320 (m/s) Light travels (much) faster (2) RA **Ignore** 'sound takes longer' or other references to time.	**(2)**

Q4.

Question number	Answer	Additional guidance	Mark
	substitution (x) =330 x 4.0 evaluation 1300 (m)	 accept 1320 (m) award full marks for correct answer without working.	**(2)** **AO2**

Q5.

Question Number	Answer	Additional guidance	Mark
	substitution (1) 0.8 =f x 4.0	(f =) $\frac{0.8}{4.0}$ allow correct substitution into seen incorrect rearrangement	(3) AO2
	rearrangement and evaluation (1) 0.2 (Hz)	award 2 marks for the correct answer with no working	
	unit (1) Hz / s^{-1}/ per sec	accept hz or hertz independent mark accept recognisable spelling	

Q6.

Question number	Answer	Additional guidance	Mark
	substitution (1) 330 = f x 0.75 rearrangement (1) (f =) $\frac{330}{0.75}$ evaluation (1) (f =) 440 (Hz)	substitution and rearrangement may be in either order. $f = \frac{v}{\lambda}$ if no other marks scored then award 1 mark for an answer that rounds to 0.0023 or 250 award full marks for the correct answer without working	**3** **AO2.1**

Q7.

Question Number	Answer	Additional guidance	Mark
	substitution (1) 300 : 1500 evaluation (1) 1:5	 $\frac{300}{1500}$ 0.2 OR $\frac{1}{5}$ ignore any units award full marks for the correct answer without working allow 1 mark for either 5:1 OR 5	**(2)**

Q8.

Question number	Answer	Additional guidance	Mark
	substitution and unit conversion (1) $470 \times 10^{-9} \times 6.30 \times 10^{14}$ answer (1) 2.96×10^{8} (m/s)	award full marks for correct numerical answer without working ecf unit conversion	**(2)**

Q9.

Question Number:	Answer	Additional guidance	Mark
	recall and substitution (1) (v =) 0.25 x 1.5 evaluation (1) 0.38 (m/s)	 accept 0.375 or 0.37 (m/s) accept 37.5, 37 or 38 for 1 mark only award full marks for the correct answer without working	**(2)**

Q10.

Question number	Answer	Additional guidance	Mark
	uses data taken from x axis (1) 28(cm) (1)	 award full marks for correct answer without working	**(2)** **AO3**